MDP Project — Review 2

What mam asked for, and what we have — at a glance

#	What mam asked	What it means	Done	Where to find it
1	**Introduction gaps written**	The research gaps, written into the Introduction	**Yes**	Gaps G1-G6 (Literature Review PDF; paper §I)
2	**Research paper on a timeline (2 timelines)**	Two timelines of the LATEST work (2024-2026): (A) prediction & causal ML, (B) source apportionment & optimization	**Yes**	Two timeline figures + tables (Literature Review PDF)
3	**The research paper**	The full, written research paper	**Yes**	MDP_Research_Paper.pdf (16 pages, LaTeX)
4	**Gaps we are trying to solve**	The specific problems our work fixes	**Yes**	Gaps G1-G6, stated plainly
5	**Our contributions & novelty**	What is new and ours, mapped to each gap	**Yes**	Contributions C1-C6 + one-line novelty statement
6	**SCI & Q1 papers to read**	Real, recent top-journal papers to read & cite	**Yes**	7 verified papers (DOIs) + a 5-member reading split

In one sentence

We built an air-quality machine-learning pipeline on real Indian data, tested it honestly, found the big causal/optimization claims were not supported, and reframed it as an **open, leakage-safe benchmark** — with a written paper, two up-to-date literature timelines, clearly stated gaps and novelty, and a verified SCI/Q1 reading list. Everything above is done and in the repository.

The only things left (not writing — decisions)

- Confirm the honest ‘benchmark’ framing with mam, and pick a target journal.
- Finalize author order (currently: Krishiykeishav, Pratyush, Shreerag, Jayshil, Ryan).
- Each member reads their assigned SCI/Q1 paper and notes which gap (G1-G6) it supports.
- Confirm one pending DOI (Qi et al. 2024) on IOPscience; confirm quartiles on Scimago.

All deliverables are on the `review/scientific-audit` branch and in `audit/review2/`: the paper, the literature review, pseudocode, tool justification, the publication plan, and figures.